MATH 2241 INTRODUCTION TO MATHEMATICAL ANALYSIS, FALL SEMESTER 2025
CHAPTER 1: REAL NUMBERS

K.Y. CHAN

Goal of this section:

(1) Real Numbers: algebraic properties and ordering
(2) Concept of upper/lower bounds, maximum/minimum, supremum/infimum
(3) Dense property of $\mathbb{Q}$
(4) Uncountability of $\mathbb{R}$ and irrational numbers

Reference books:

- Robert G. Bartle, Donald R. Sherbert: Introduction to Real Analysis (Wiley, 2011, Fourth Edition)
- Kenneth A. Ross: Elementary Analysis: The Theory of Calculus (Springer, 2013, Second Edition)

0.1. **Overview of the course.** This course is to rebuild your understanding of some previous courses. For example, we emphasis on how to define real numbers, differentiations and integrations from foundations and axioms. Such abstract understanding is important to build different mathematical theory for other contents and applications in other realms.

1. Basic terminology and property

1.1. Fields and their algebraic properties.

Definition 1.1. A field is a set with two binary operations, called addition $+ : F \times F \to F$ and multiplication $\cdot : F \times F \to F$ on F such that the following properties hold:

(A1) (commutativity for addition) $a + b = b + a$ for $a, b \in F$
(A2) (associativity for addition) $(a + b) + c = a + (b + c)$;
(A3) (existence of a zero element) there exists an element 0 in F such that $0+a = a + 0 = a$ for all $a \in F$;
(A4) (existence of negatives) for each $a \in F$, there exists an element $-a$ in F such that $a + (-a) = (-a) + a = 0$;

(M1) (commutativity for multiplication) $a \cdot b = b \cdot a$ for $a, b \in F$;
(M2) (associativity for multiplication) $a \cdot (b \cdot c) = (a \cdot b) \cdot c$;
(M3) (existence of an identity element) there exists an element 1 in F not equal to 0 such that $1 \cdot a = a \cdot 1 = a$;
(M4) (existence of multiplicative inverses) for each $a \neq 0$ in F, there exists an element $1/a \in F$ such that $a \cdot (1/a) = (1/a) \cdot a = 1$;
(D) (distributive property of addition over multiplication) $a \cdot (b+c) = a \cdot b + a \cdot c$ for all $a, b, c \in F$.

Exercise 1.2. Show that

(a) the element 0 is unique i.e. if $0'$ is also an element in F such that $0' + a = a + 0' = a$ for all a, then $0' = 0$.
(b) the negative $-a$ in (A4) is unique.
(c) the multiplicative identity 1 is unique
(d) the element $1/a$ in (M4) is unique.

Remark 1.3. (Subtraction) For $a, b \in F$, $a - b$ means $a + (-b)$.

Remark 1.4. (Division) For $a, b \in F$ with $b \neq 0$, $\frac{a}{b}$ means $a \cdot (1/b)$. We may also sometimes write a^{-1} for $\frac{1}{a}$.

We now deduce some more properties:

Proposition 1.5. *Let F be a field.*

(a) *If z and a are elements in F with $z + a = a$, then $z = 0$.*
(b) *If u and $b \neq 0$ are elements in F with $ub = b$, then $u = 1$.*
(c) *If $a \in F$, then $0a = 0$.*
(d) *If $a \in F$, then $-(-a) = a$.*
(e) *If $a, b \in F$, then $(-a)(b) = -ab$.*
(f) *If $a, b \in F$, then $(-a)(-b) = ab$.*

Proof. For (a), $z + a = a \Rightarrow z + a + (-a) = a + (-a) \Rightarrow z = 0$
For (b), $ub = b \Rightarrow ub \cdot \frac{1}{b} = b \cdot \frac{1}{b} \Rightarrow u = 1$
For (c), $(0 + 0)a = 0$ and so $0a + 0a = 0a$. This then follows from (a).
For (d), according to definition, to show $-(-a) = a$, it is to check $(-a) + a = 0$, but this follows from (A4).

For (e), this follows from:

$$\begin{aligned}(-a)(-b) &= -(a(-b)) \quad \text{by (e)}\\ &= -((-b)a) \quad \text{by (M1)}\\ &= -(-ba) \quad \text{by (e)}\\ &= ba \quad \text{by (d)}\\ &= ab \quad \text{by (M1)}\end{aligned}$$

□

1.2. **Ordered fields.**

Definition 1.6. A field F is said to be ordered if there exists an ordering structure $\leq$ satisfying the following properties:

(O1) For any $a, b \in F$, $a \leq b$ or $b \leq a$.
(O2) For $a, b \in F$, if $a \leq b$ and $b \leq a$, then $a = b$.
(O3) For $a, b, c \in F$, if $a \leq b$ and $b \leq c$, then $a \leq c$.
(O4) For $a, b, c \in F$, if $a \leq b$, then $a + c \leq b + c$.
(O5) For $a, b, c \in F$, if $a \leq b$ and $c \geq 0$, then $ac \leq bc$.

For an ordered field F, we shall write $a < b$ if $a \leq b$ and $a \neq b$.

The typical example of an ordered field F is $\mathbb{Q}$.

Exercise 1.7. If F is an ordered field, deduce the followings from the definitions of an ordered field:

(a) If $a \neq 0$, then either $0 < a$ or $0 < -a$.
(b) For any $a \in F$ and $a \neq 0$, $a^2 > 0$.
(c) $0 < 1$.
(d) If $0 < a$, then $0 < a^{-1}$.
(e) For any $a, b \in F$, if $a < b$, then $-b < -a$.
(f) For any $a, b \in F$, if $a < b$ and $c > 0$, then $ac < bc$.
(g) For any $a, b \in F$, if $0 < a$ and $0 < b$, then $a < b$ if and only if $a^2 < b^2$.
(h) $1 + 1 \neq 0$.

1.3. **Some examples.**

Definition 1.8.
(1) The set $\mathbb{N}$ of natural numbers $= \{1, 2, 3, \ldots\}$. (The existence of $\mathbb{N}$ is from Peano's axiom.)
(2) The set $\mathbb{Z}$ of integers$=\{\ldots, -2, -1, 0, 1, 2, \ldots\}$.
(3) The set $\mathbb{Q}$ of rational numbers$=\left\{\frac{p}{q} : pq \in \mathbb{Z}, q \neq 0\right\}$.

The order structure on $\mathbb{Q}$ is defined as: for any $n \in \mathbb{N}$, $0 < n$, and for any $\frac{p_1}{q_1}, \frac{p_2}{q_2} \in \mathbb{Q}$, $\frac{p_1}{q_1} < \frac{p_2}{q_2}$ if and only if $q_1 p_2 - p_1 q_2 \in \mathbb{N}$.

Exercise 1.9. Write down a relation between the sets $\mathbb{Z}, \mathbb{N}, \mathbb{Q}$.

Exercise 1.10. The field $\mathbb{Q}[\sqrt{-1}]$ is defined as the set of numbers of the form $a+bi$, where i is just a symbol called an imaginary number, and $a, b \in \mathbb{Q}$. Define

- (addition) $(a_1 + b_1 i) + (a_2 + b_1 i) = (a_1 + a_2) + (b_1 + b_2)i$
- (multiplication) $(a_1 + b_1 i) \cdot (a_2 + b_2 i) = (a_1 a_2 - b_1 b_2) + (a_1 b_2 + b_1 a_2)i$.

Check $\mathbb{Q}[\sqrt{-1}]$ is a field. Show that $\mathbb{Q}[\sqrt{-1}]$ is not ordered.

2. Maximum and minimum

2.1. Upper bound and lower bound.

Definition 2.1. Let S be a non-empty subset of an ordered field.

(1) The set S is said to be bounded above if there exists a real number M such that $x \leq M$ for all $x \in S$. We say that M is an upper bound of S.
(2) The set S is said to be bounded below if there exists a real number m such that $m \leq x$ for all $x \in S$. We say that m is a lower bound of S.
(3) The set S is said to be bounded if S is both bounded above and bounded below.

Example 2.2. (1) Let $S_1 = \{x \in \mathbb{Q} : -1 < x \leq 2\} \subset \mathbb{Q}$. Then 2 and 3 are upper bounds of S_1 and indeed any number equal to or greater than 2 is also an upper bound of S. Thus S_1 is bounded above. Similarly, any number equal to or less than -1 is a lower bound of S_1, and so S_1 is bounded below. Thus, S_1 is bounded.
(2) Let $S_2 = \{x \in \mathbb{Q} : x < 3\} \subset \mathbb{Q}$. Then S_2 is bounded above but not bounded below.

2.2. Maximum and minimum.

Definition 2.3. Let S be a non-empty subset of an ordered field F.

(1) The *maximum* of S is an element s_0 in S such that $x \leq s_0$ for all $x \in S$.
(2) The *minimum* of S is an element i_0 in S such that $i_0 \leq x$ for all $x \in S$.

Remark 2.4. (1) If S is not bounded above, then the maximum does not exist.
(2) However, even if S is bounded above, the maximum may not exist in general. For example, the set

$$\{x \in \mathbb{Q} : 0 \leq x < 8\}$$

does not have a maximum.

(3) Similarly, if S is not bounded below, then the minimum does not exist.
(4) Even if S is bounded below, the minimum may not exist in general.
(5) Note that if a maximum/minimum of S exists, it is unique (why?).

Example 2.5. Determine whether the following sets have maximum and/or minimum. If it/they exist, find it/them.

(1) $S_1 = \{x \in \mathbb{Q} : -1 < x \leq 2\}$
(2) $S_2 = \{x \in \mathbb{Q} : -2 < x \leq 1 \text{ or } 2 < x \leq 5\}$
(3) $S_3 = \{x \in \mathbb{Q} : -2 \leq x\}$
(4) $S_4 = \{x \in \mathbb{Q} : x^3 = 1\}$
(5) $S_5 = \{x \in \mathbb{Q} : x^3 < 1\}$
(6) $S_6 = \mathbb{N}$
(7) $S_7 = \mathbb{Q}$

3. Completeness property

3.1. Supremum and infimum.

Definition 3.1. Let S be a non-empty subset of an ordered field F.

(1) If S is bounded above (i.e. an uper bound exists), then the supremum, denoted by $\sup S$, of S is the least upper bound (in F) of S. In other words, $\sup(S)$ is the minimum of the set

$$\{u \in F : u \text{ is an upper bound of } S\}.$$

(2) If S is bound below (i.e. a lower bound exists), then the infimum, denoted by $\inf S$, of S is the greatest lower bound (in F) of S. In other words, $\inf(S)$ is the maximum of the set

$$\{u \in F : u \text{ is a lower bound of } S\}.$$

Remark 3.2. (1) In contrast with maximum/minimum, the supremum/infimum of a non-empty S (if exists) may not be in S.
(2) It the supremum/infimum exists, then it is unique.

Proposition 3.3. *(Criteria for finding supremum) Let F be an ordered field. An upper bound u of a non-empty set S is the supremum if and only if for any given $\epsilon > 0$, there exists an $x \in S$ such that $u - x < \epsilon$.*

Proof. Only if direction:
Suppose u is the supremum. Suppose there exists $\epsilon' > 0$ such that $u - x \geq \epsilon'$ for all $x \in S$, then $x \leq u - \epsilon'$ for all $x \in S$. Then $u - \epsilon'$ is also an upper bound for S. This contradicts the definition of the supremum.

If direction:
Suppose u' is also an upper bound of S. Let $\epsilon = u - u'$. If $u > u'$, then there exists $x \in S$ such that $u - x < \epsilon$. Hence, $u - x < u - u'$ and so $u' < x$. This contradicts that u' is an upper bound. Hence, we must have $u \leq u'$, and so u is the least upper bound i.e. u is the supremum. □

Exercise 3.4. Write down the supremum and infimum of the sets S_1 to S_6 in Example 2.5 (if exists).

Exercise 3.5. (a) Formulate a statement analogous to Proposition 3.3 that uses inf rather than sup.
(b) Write down $\inf\left\{\frac{1}{n} : n \in \mathbb{N}\right\}$.

Exercise 3.6. Let S be a bounded above non-empty subset of an ordered field F. Suppose $\sup(S)$ exists. Let $S' = \{-s : s \in S\}$. Then $\inf S' = -\sup S$.

Exercise 3.7. Let S and T be two bounded above non-empty subsets of an ordered field F. Suppose $\sup(S)$ and $\sup(T)$ exist in F. Define

$$S + T := \{s + t : s \in S, t \in T\}.$$

Show that

$$\sup(S + T) = \sup S + \sup T.$$

3.2. **Completeness property.** The completeness property refers to the following fact:

Definition 3.8. An ordered field F is said to be *complete* (or satisfy the completeness axiom) if for any bounded-above non-empty subset S of F, the supremum of S exists.

Definition 3.9. Define the field $\mathbb{R}$ of real numbers to be the complete ordered field.

Such field exists and is unique, and contains $\mathbb{Z}, \mathbb{N}, \mathbb{Q}$ as (order-preserving) subsets. (We shall not go into mathematics detail on these.) $\mathbb{Q}$ is not complete and we shall explore more in next section (which we need more preparations).

Exercise 3.10. Use the completeness property to deduce that: for any non-empty subset S of $\mathbb{R}$, if S is bounded below, then the infimum exists.

4. Denseness property of $\mathbb{Q}$

4.1. **Well-orderedness property of $\mathbb{N}$.**

Theorem 4.1. *(Well-ordered property of $\mathbb{N}$) Any non-empty subset of $\mathbb{N}$ has a minimum.*

Proof. Let S be a non-empty subset of $\mathbb{N}$. Let $n \in S$. We find the least integer (under the order $\leq$) in $\{1, \ldots, n\}$ that is in S. Then it is the minimum of S. □

Lemma 4.2. *The set $\mathbb{N}$ is not bounded above in $\mathbb{R}$.*

Proof. Suppose $\mathbb{N}$ is bounded above. Then let $s = \sup(\mathbb{N})$. Then $s - 1$ is not an upper bound. Thus there exists $n' \in \mathbb{N}$ such that $s - 1 < n'$. Hence, $s < n' + 1$. This contradicts that s is an upper bound of $\mathbb{N}$ □

Corollary 4.3. *Let $x \in \mathbb{R}$ with $x > 0$. Then there exists $n_x \in \mathbb{N}$ such that $n_x - 1 \leq x < n_x$.*

Proof. Let $S = \{y \in \mathbb{N} : x < y\}$. By Lemma 4.2, $S \neq \emptyset$. By the well-ordered property, let n be the minimum element of S. Then $n - 1$ cannot be in S and so $n - 1 \leq x$. Thus we have $n - 1 \leq x < n$. □

Exercise 4.4. For $a, b \in \mathbb{R}$, if $a, b > 0$, then there exists an integer $n \in \mathbb{N}$ such that $b < an$.

Corollary 4.5. *Let $x \in \mathbb{R}$. If $0 < x$, then there exists an integer $n \in \mathbb{N}$ such that $\frac{1}{n} < x$.*

Proof. Since $0 < x$, $0 < \frac{1}{x}$. By Exercise 4.4, $\frac{1}{x} < n$ for some integer $n \in \mathbb{N}$. Hence, $\frac{1}{n} < x$. □

Corollary 4.6. *(Denseness property of $\mathbb{Q}$) Let $a, b \in \mathbb{R}$. Suppose $a < b$. Then there exists a rational number r such that $a < r < b$.*

Proof. We may assume that $0 < a < b$. By Corollary 4.5,

$$\frac{1}{n} < b - a$$

for some $n \in \mathbb{N}$. Then $na + 1 < nb$. By Corollary 4.3,

$$m - 1 \leq na < m$$

and so $m \leq na + 1 < nb$. Combining, we have:

$$na < m < nb$$

and so $a < \frac{m}{n} < b$. □

4.2. **Existence of irrational numbers.**

Definition 4.7. A number a in $\mathbb{R}$ is said to be irrational if a is in $\mathbb{Q}$ i.e. $a \notin \mathbb{Q}$.

Theorem 4.8. *There is no rational number x such that $x^2 = 2$.*

Proof. Suppose x is a rational number such that $x^2 = 2$. Write $x = \frac{p}{q}$ for some non-zero integers p and q. We may and shall assume that p and q have no common factors.

Then $\left(\frac{p}{q}\right)^2 = 2$ and so $p^2 = 2q^2$. Thus p must be even and so $p = 2r$ for some integer r. Hence, $4r^2 = 2q^2$ and so $q^2 = 2r^2$. Then q must be even. However, this implies that p and q have a common factor. This leads to a contradiction. □

Theorem 4.9. *(Existence of square root 2) There exists $x \in \mathbb{R}$ such that $x^2 = 2$. Moreover $x = \sup(\{y \in \mathbb{R} : y^2 \leq 2\})$ (By Theorem 4.8, x is irrational.)*

Proof. Let

$$S = \left\{y \in \mathbb{R} : y^2 \leq 2\right\}.$$

The set is bounded above. Then the axiom implies that sup S exists. Let $s = \sup S$. We shall show that $s^2 = 2$.

Case 1: $s^2 < 2$. We shall show that there exists $n \in \mathbb{N}$ such that $\left(s + \frac{1}{n}\right)^2 < 2$. This will then contradict that s is an upper bound.

By Corollary 4.5, there exists $m \in \mathbb{N}$ such that

$$s^2 + \frac{1}{m} < 2.$$

Now one choose n' such that, by Corollary 4.5 again,

$$\frac{1}{n'} < \frac{1}{(2s+1)m}$$

and so

$$\left(s + \frac{1}{n'}\right)^2 < s^2 + \frac{2s+1}{n'} < s^2 + \frac{1}{m} \leq 2.$$

Thus This gives the desired contradiction.

Case 2: $s^2 > 2$. By solving some suitable equations as above (exercise, or see the details on a textbook), one can find $n \in \mathbb{N}$ such that $2 < \left(s - \frac{1}{n''}\right)^2$. This implies that $x < s - \frac{1}{n''}$ for all $x \in S$, and so contradicts that s is the least upper bound of S.

Since Case 1 and Case 2 cannot happen, we must have that $s^2 = 2$. □

Exercise 4.10. Explain from axioms that there exists a number $x \in \mathbb{R}$ such that $x^2 = 2$ and $x > 0$.

The following exercises show that the set of irrational numbers also satisfies the denseness property.

Exercise 4.11. (Dense property of irrational numbers) Show that for any numbers $a, b \in \mathbb{R}$, if $a < b$, there exists an irrational number t such that $a < t < b$.

4.3. $\mathbb{Q}$ **is not complete.**

Example 4.12. Let $S = \{x \in \mathbb{Q} : x^2 < 2\}$. There is *no* number s in $\mathbb{Q}$ such that s is the least upper bound of S.
Proof: Let s be an arbitrary upper bound of S in $\mathbb{Q}$. Let $s' = \sup(S)$. By Theorems 4.8 and 4.9, $s' \notin \mathbb{Q}$ and so $s \neq s'$. Thus $s' < s$. By denseness of $\mathbb{Q}$, there exists $r \in \mathbb{Q}$ such that $s' < r < s$. Then r is also an upper bound of S and is in $\mathbb{Q}$. Hence, any arbitrary upper bound of S cannot be the least one in $\mathbb{Q}$.

Corollary 4.13. *The ordered field* $\mathbb{Q}$ *is not complete.*

5. Uncountability of $\mathbb{R}$

In this section, we shall prove that $\mathbb{R}$ is uncountable. The countability is related to the sequence in the next chapter. The method we prove is to use nested sequences. We shall first introduce some notations.

5.1. **Some notations and absolute values.** From now on, we shall only consider $F = \mathbb{R}$.

Definition 5.1. For any $a \in \mathbb{R}$, define

$$|a| = a \quad \text{if } a \geq 0 \quad \text{or} \quad |a| = -a \quad \text{if } a < 0 \, .$$

Exercise 5.2. For $a, b \in \mathbb{R}$, show that $|a + b| \leq |a| + |b|$.

5.2. **Intervals.** For $a < b$, we shall write

$$(a, b) := \{x \in \mathbb{R} : a < x < b\} \quad \text{(open bounded interval)},$$

$$[a, b) := \{x \in \mathbb{R} : a \leq x < b\} \, ,$$

$$(a, b] := \{x \in \mathbb{R} : a < x \leq b\} \, ,$$

$$[a, b] := \{x \in \mathbb{R} : a \leq x \leq b\} \quad \text{(closed bounded interval)}.$$

We shall use the symbols ∞ and $-\infty$, which are called *infinity* and *negative infinity*. Define

$$(-\infty, a] := \{x \in \mathbb{R} : x \leq a\} \, , \quad (-\infty, a) := \{x \in \mathbb{R} : x < a\} \, ,$$

$$[a, \infty) := \{x \in \mathbb{R} : x \geq a\} \, , \quad (a, \infty) := \{x \in \mathbb{R} : x > a\} \, .$$

Exercise 5.3. Write the following sets in interval notations:

(1) $\{x \in \mathbb{R} : x^2 < 4\}$
(2) $\{-x^2 \in \mathbb{R} : x \in \mathbb{R}\}$

5.3. **Nested sequences.**

Definition 5.4. A sequence of intervals I_n $(n \in \mathbb{N})$ in is said to be nested if the following chain of inclusions

$$I_1 \supseteq I_2 \supseteq I_3 \supseteq \dots$$

holds.

Example 5.5.
(1) Let $I_n = [0, \frac{1}{n})$. Then I_n forms a nested sequence of intervals. Moreover, $\cap_n I_n = \{0\}$.
(2) Let $J_n = (0, \frac{1}{n})$. Then J_n also forms a nested sequence of intervals. Moreover, $\cap_n J_n = \emptyset$.
(3) Let $K_n = [n, \infty)$. Then K_n forms a nested sequence of intervals. Moreover, $\cap_n K_n = \emptyset$.

Exercise 5.6. Work out a detail argument of the above intersections. Point out which key theorem is needed in shoing the above intersections.

Theorem 5.7. *(Nested interval property) Let $I_n = [a_n, b_n]$ $(n \in \mathbb{N}, a_n, b_n \in \mathbb{R})$ be bounded closed intervals. Suppose I_n $(n \in \mathbb{N})$ form a nested sequence of intervals. Then $\cap_n I_n \neq \emptyset$.*

Proof. Let $S = \{a_n : n \in \mathbb{N}\}$. Let $s = \sup S$.

Step 1: Show each b_n is an upper bound of S.
Fix an arbitrary $n^* \in \mathbb{N}$. If $n \leq n^*$, then $a_n \leq a_{n^*} \leq b_{n^*}$. If $n > n^*$, then $a_n \leq b_n \leq b_{n^*}$. Combining the two, we have $a_n \leq b_{n^*}$ for all $n \in \mathbb{N}$. Hence, b_{n^*} is an upper bound of S.

Step 2: Show that $s \leq b_n$ for all n.
Since s is the least upper bound for S, Step 1 implies that $s \leq b_n$ for all n.

Step 3: Show $a_n \leq s$ for any $n \in \mathbb{N}$.
Indeed, $a_n \leq s$ for all $n \in \mathbb{N}$ follows from that s is an upper bound of S.

Step 4: Conclusion
For any $n \in \mathbb{N}$, Step 3 implies that $a_n \leq s$ and Step 2 implies that $s \leq b_n$. Hence, $s \in I_n = [a_n, b_n]$ for all $n \in \mathbb{N}$. Thus $s \in \cap_{n \in \mathbb{N}} I_n$. □

5.4. **Application: Uncountability of real numbers.** We recall the definition of a countable set.

Definition 5.8. (1) A set S is said to be countably infinite if there exists a bijection from $\mathbb{N}$ to S.
(2) A set S is said to be countable if S is either finite or countably infinite.
(3) A set S is said to be uncountable if S is not countable.

Lemma 5.9. *Any subset of a countable set is still countable.*

Proof. Let S be a subset of a countable set. If S is finite, then we are done. Suppose S is infinite. It suffices to consider that S is an infinite subset of $\mathbb{N}$. We define a map from N to S as follows: $f(1) = \min(S)$ and recursively, define $f(n) = \min(S \backslash \{f(1), \ldots, f(n-1)\})$. It is well-defined since $S \backslash \{f(1), \ldots, f(n-1)\}$ is non-empty and from Theorem 4.1.

Since $f(i) < f(j)$ for $i < j$, we have $f(i) \neq f(j)$ for $i \neq j$. This implies that f is injective.

We now show f is surjective. Let $m \in S$. For any $i \in S \setminus \{f(1), \ldots, f(m)\}$, we have $f(1) < \ldots < f(m) < i$. Thus we have $m < i$. This implies that $m \notin S \setminus \{f(1), \ldots, f(m)\}$ and so in $n \in \{f(1), \ldots, f(m)\}$ i.e. $f(n) = m$ for some $n \in \{1, \ldots, m\}$. Thus f is surjective. □

Theorem 5.10. *The set $\mathbb{R}$ of real numbers is uncountable.*

Proof. Let $I = [0, 1]$. We shall show that $[0, 1]$ is uncountable. Since $[0, 1]$ is a subset of $\mathbb{R}$, this will imply $\mathbb{R}$ is also uncountable by Lemma 5.9.

Suppose $[0, 1]$ is countable to derive a contradiction. Then enumerate $I = \{x_1, x_2, x_3, \ldots\}$. By Corollary 4.6, we inductively construct a closed interval I_n such that I_n is a subset of I_{n-1} and $x_n \notin I_n$. By Theorem 5.7, $\cap_{n\in\mathbb{N}} I_n \neq \emptyset$. Since $x_n \notin I_n$ and so $x_n \notin \cap_{n\in\mathbb{N}} I_n$. Hence $\zeta \neq x_n$ for all n. This gives a contradiction. Thus $[0, 1]$ is uncountable. □

5.5. **Countably infinite of rational numbers.**

Proposition 5.11. *The set $\mathbb{Z}$ is countable.*

Proof. Define $f : \mathbb{Z} \to \mathbb{N}$ by

$$f(n) = \begin{cases} 2n & \text{if } n > 0 \\ 2(-n) + 1 & \text{if } n \leq 0 \end{cases}$$

One checks that f is bijective. Thus $\mathbb{Z}$ is countable by definition. □

Proposition 5.12. *The Catersian product of two countable sets is countably infinite.*

Proof. Define $f : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ by

$$f(n_1, n_2) = 2^{n_1} 3^{n_2}.$$

By the unique prime factorization, the map f is injective. By Lemma 5.9, $f(\mathbb{N}\times\mathbb{N})$ is countably infintie and so $\mathbb{N}\times\mathbb{N}$ is also countably infinite. □

Proposition 5.13. *The set $\mathbb{Q}$ is countable.*

Proof. Define $f:\mathbb{Q}\to\mathbb{Z}\times\mathbb{N}$ given by: for any $r\in\mathbb{Q}$, $r=\frac{p}{q}$ for some unique relative prime integers p,q with $q>0$, and so one assigns a map:

$$f(\frac{p}{q})=(p,q).$$

Since $\mathbb{Z}$ is countably infinite, $\mathbb{N}\times\mathbb{Z}$ is countably infinite by Proposition 5.12. Thus, by Lemma 5.9, $f(\mathbb{Q})$ is countably infinite. Since f is injective, $\mathbb{Q}$ is then also cuntably infinite. □

5.6. **Irrational numbers.**

Corollary 5.14. *The set of irrational numbers is uncountable.*

Proof. The union of two countable sets is still countable. Since the union of the set of rational numbers and the set of irrational numbers is uncountable, and the set of rational numbers is countable, we must have $\mathbb{R}$ is uncountable. □